Week 7 Level 1: Introduction

Hello, Python!

Text code is just blocks written as words — and Python is picky.

What you will learn

- How to open, run and save a program in **Thonny**.
- How to make Python say things with `print`, and leave notes with comments.
- How to make the turtle draw a square with `forward` and `left`.
- How to read a red error message like a detective.

Warm-up: Spot the difference

Python is very picky. On each card, one line works and one line confuses Python. Circle the difference. Check the line that works.

Card	Line A	Line B
1	☐ `print("Hello")`	☐ `Print("Hello")`
2	☐ `t.forwad(100)`	☐ `t.forward(100)`
3	☐ `t.left(90`	☐ `t.left(90)`
4	☐ `t.left(90)`	☐ `t.Left(90)`
5	☐ `print(Hi there)`	☐ `print("Hi there")`

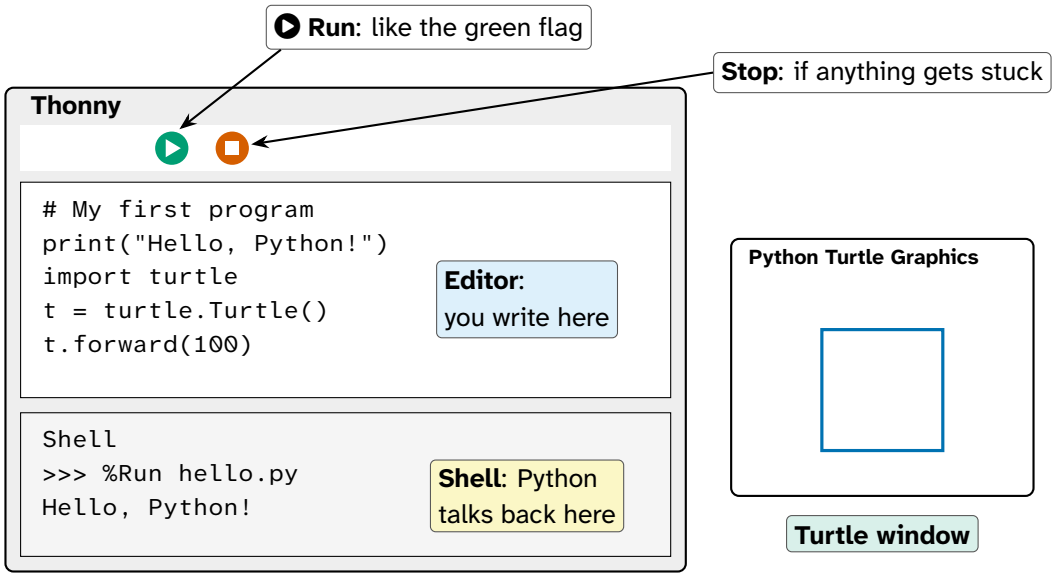
The big idea: blocks become words

In Scratch you snapped blocks together. In Python you **type** the same ideas as words. The computer still does the lines in order, from top to bottom.

Scratch block		Python line
when clicked	→	the green **Run** button (or F5)
move 100 steps	→	`t.forward(100)`
turn 90 degrees	→	`t.left(90)`
say Hello!	→	`print("Hello!")`

Meet Thonny

Thonny is the app where we write Python. It has three important parts.



Tip: Where did my turtle go?

The turtle window sometimes hides behind Thonny. Click the Python icon in the Dock at the bottom of the screen.

Every turtle program has the same shape

```
import turtle              # 1. get the turtle toolbox
t = turtle.Turtle()        # 2. make a turtle and call it t
t.forward(100)             # 3. your drawing commands go here
t.left(90)
turtle.done()              # 4. keep the window open at the end
```

Lines that start with # are **comments**. Python skips them. They are notes for people.

Exercise 7.1: Hello, Python!

Folder: `week07/ex-hello-python`. Open `starter.py` and press Run.

1. Add three `print` lines: who is coding today (a **nickname**), your favorite game or animal, and a goodbye that uses the word `bye`.
2. Press Run after every new line.

Done looks like: four different lines in the Shell. Check: open `check_my_work.py` and press Run.

Exercise 7.2: My First Square

Folder: `week07/ex-first-square`. The turtle draws one side already. Add the other three sides: each side is `t.forward(100)` then `t.left(90)`. Done looks like: a closed square — the turtle ends where it started. Then run `check_my_work.py`.

Reading a red error message

When Python is confused, it stops and shows red text in the Shell. Don't panic — it is a clue!

Output

```
Traceback (most recent call last):
  File "starter.py", line 12, in <module>
    Print("Typo hunt: fixed!")
NameError: name 'Print' is not defined
```

- **line 12** tells you **where** to look.
- The **last line** tells you **what** confused Python: it does not know the word `Print`.

Exercise 7.3: Typo Hunt

Folder: `week07/ex-typo-hunt`. This program should draw a flag on a pole, but it has **4 typos**. Run it, read the red message, fix **one** typo, run again. Write down what you fixed:

Line	The typo was	I changed it to

Tip: Save your work safely

Exercises: press Cmd+S to save `starter.py` in its folder. New programs go in `Documents/my-python`. Use names that say what the program does, like `flag-drawing.py`. Never call a file `turtle.py`: Python would get mixed up with the real turtle!

Vocabulary box

Python A text programming language. We type words instead of snapping blocks.

Thonny The app where we write and run Python programs.

Editor The top part of Thonny where you write your program.

Shell The bottom part of Thonny where Python shows messages and errors.

print A command that shows words in the Shell: `print("Hi")`.

Comment A note for people that starts with #. Python skips it.

Syntax error A typo or missing bracket that stops Python from understanding your code.

Bug A mistake in a program. Every programmer makes them!

Reflection

What was different about typing code instead of dragging blocks?

Which typo was the hardest to spot? What will you check first next time?

Week 7
Triangle Time

Level 1: Introduction

15–20 minutes. Do Core first; Challenge and Creative are optional.

Name (nickname is fine): _____ Date: _____

★ Core (everyone): Triangle Time (about 10 minutes)

In Thonny, open `week07/hw-triangle/starter.py`. The turtle already draws **one** side.

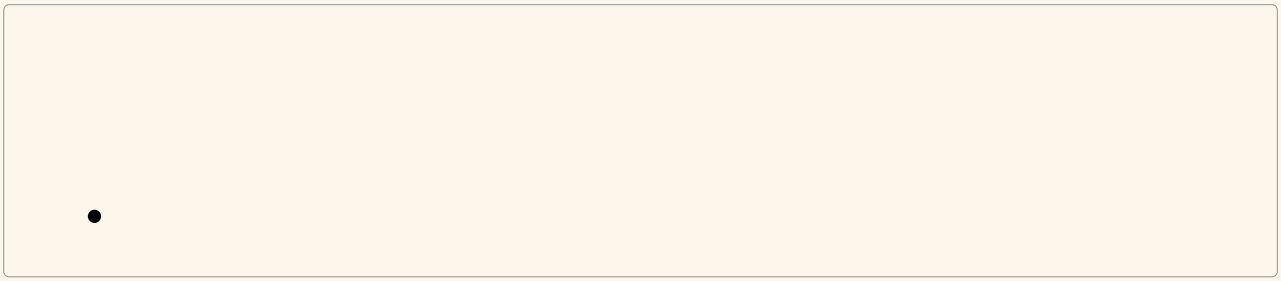
1. Add the code for the other two sides. Each side is `t.forward(120)` then `t.left(120)`.
2. Why 120? The turtle must turn all the way round (360 degrees) in 3 corners: $3 \times 120 = 360$.
3. Press Run. Then open `check_my_work.py` and press Run.

No computer at home? Write the whole program here and type it in next week:

★★ Challenge (want more?): Be the turtle

Don't run it yet! Read this program and draw what the turtle will draw in the box. The turtle starts at the dot, facing right. Then type it in and check.

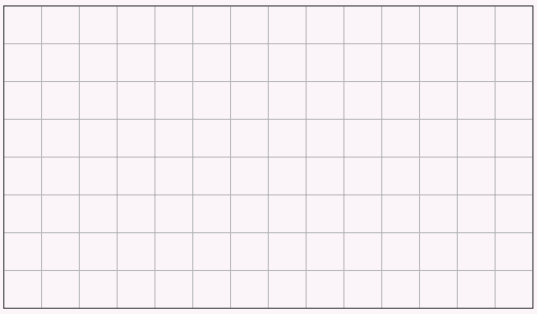
```
t.forward(100)
t.left(90)
t.forward(50)
t.left(90)
t.forward(100)
```



What letter or shape does it look like? _____

Creative (make it yours): My initial

Plan the first letter of your nickname with straight lines only (E, F, H, L, T, Z work well). Draw it on the grid. Write the steps next to it, like `F 100, L 90, F 50`. Then try it in Thonny in a new file saved as `my-python/initial.py`.



My steps:

Self-check: tick each box when it is true

- ☐ My triangle has 3 equal sides and is closed.
- ☐ I ran `check_my_work.py` (or wrote my code on paper).
- ☐ I saved my file with a sensible name in the right folder.
- ☐ If I met a red error, I read the line number and the last line.